PRODUCT & ENGINEERING BLUEPRINT

AI Workflow SaaS Implementation Guide

From concierge-first WhatsApp pilots to a pack-based business automation platform.

Prepared for ViroByte

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31 sections | 8 phases | Scored discovery first

SELL THE OUTCOME FIRST. BUILD THE REPEATABLE CORE SECOND.

Controlled outcomes. Local-first onboarding. Proof in every report.
Packs before platform.

Planning assumptions are marked. Provider and market references are linked.
Every phase includes a completion gate.

GUIDE MAP

Guide map

Read sections 01-08 for product, market and scope; 09-15 for workflows, pricing and pilots; 16-22 for technical design; 23-31 for delivery, growth and references.

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Product definition and promise

Decision

- Build a request-to-resolution product for small and midsize teams: capture each incoming request, collect missing information, assign an owner, follow up within approved rules, and show evidence of what was completed.
- **Working descriptor:** AI Workflow SaaS - Request to Resolution. This is a product direction, not a cleared brand name.
- This is an independent ViroByte product with its own codebase, tenant data, branding, pricing and accounts. It does not depend on SchoolDesk or any other ViroByte product.

The customer promise

- Every incoming request becomes tracked work with an owner, a due date and a visible next action. Supported routine steps run automatically; exceptions go to a person with full context.
- The value is fewer forgotten requests and less manual chasing across the tools a business already uses. It is not an AI that runs the business.

How the product is built

- **Work item:** the primary object. It can be a sales inquiry, quote request, repair intake, onboarding checklist, service case or internal approval. It is not a chat thread or a calendar booking.
- **Four layers:** a shared core (identity, state, deadlines, permissions, receipts, reporting); process packs (business rules); industry presets (vocabulary, fields, approved knowledge); connectors (supported read and write actions).
- **Delivery:** assisted and paid first. The team learns from real operations before building a broad platform.

What broad applicability means

- Many industries share processes, but no first product serves every business, system or regulation. Publish the supported processes, connectors and limits.
- Large companies may later use it for one department workflow, after enterprise requirements are funded and verified.

Success criterion

- A customer renews because a supported process finishes with less effort or fewer missed steps, and subscription revenue covers usage, support and a share of operating cost. Differentiation and profitability are hypotheses until pilots prove them.

How V4 combines V2 and V3

Taken from V3 (strategy and honesty)

- Work-item core with completion contracts and an evidence ledger; segment chosen by scored interviews; one pack and one language first; setup fee plus subscription; unit economics with break-even; claims discipline; expansion gates.

Taken from V2 (build detail and ViroByte fit)

- Concrete data model, API list, queue and state design; Roman Urdu, Urdu and English language handling; local-first onboarding and support on WhatsApp; named iterations with acceptance tests; launch playbook; test plan.

V4 decisions that differ from both

Topic	V2	V3	V4
First segment	Repair shops assumed	Quote-driven firms recommended	Scored interviews across both; decide in week 2
Pricing	Low local ranges, thin margin	USD only	USD tier plus local PKR tier, each with its own value test
Team	Three-person plan	Solo plan	Three founders, roles named, shorter but gated schedule
Competitors	AI agents, chatbots	Adds CRM and workflow tools only briefly	Dedicated overlap section with CRM, helpdesk and task tools
Calendar and Sheets	In MVP	Optional	Optional, released only on paid demand

Claims removed and kept out

- No WhatsApp 'conversation fee' framing, no third-hand pricing-change claims, no 'never miss a message', no competitor price comparisons from non-official sources. Competitor facts below come from official pages checked on 10 October 2026.
- Prices, schedules and thresholds are planning assumptions. Replace them with pilot data.

What already exists and what we test

What exists today

- Respond.io advertises AI agents that qualify leads, book appointments, follow guardrails and hand off to humans [R1]. Zapier Agents run delegated work across 9,000+ connected apps [R2]. n8n documents AI-enabled workflows [R3].
- Grok Bot (xAI, beta, launched 11 August 2026) is an owner-side personal agent that works on its own cloud computer [R11]. It helps one person do tasks. It is not a governed, customer-facing, multi-user process with owners, deadlines and evidence.
- **Conclusion:** AI replies, connectors and workflow templates are not a unique claim. Do not claim to be first, only or better than any named product.

The distinction we will test

- A packaged follow-through service for one narrow process: every eligible request has a responsible person, a deadline, a next step and completion evidence. A queue shows stalled work; permitted reminders and escalations move it forward; a weekly report links results to individual items.

Four things to prove in pilots

- **Completion contract:** each pack states trigger, required information, allowed actions, owner, deadline, escalation and acceptable evidence.
- **Evidence-backed operation:** a message sent, a request acknowledged, a provider-confirmed action and a customer outcome are different events and are never shown as the same thing.
- **Exception learning:** repeated failure reasons become reviewed improvements to shared pack rules and tests. Staff corrections never silently change live behavior.
- **Fast local setup:** Roman Urdu, Urdu and English handling, local prices and WhatsApp-based support. Setup time falls as similar customers are added.

Competitive test

- Ask five qualified buyers to compare our process with their current tools and one relevant platform, after seeing total cost and setup effort. If most prefer the alternative, narrow the process or change the offer before building more.
- Advantage can accumulate through tested packs, reliable connectors and onboarding know-how. These are execution advantages, not patents or an uncopiable moat.

CRM, helpdesk and task tools

Why this is the closest competition

- A CRM (deals and follow-up tasks), a helpdesk (tickets, SLAs, assignment) or a task tool (owners and due dates) can already track a request to resolution. Many small businesses use a spreadsheet or WhatsApp itself.
- The risk is not that these tools are better AI. It is that the buyer says 'our CRM can do that' and the setup effort is lower than our fee.

Where we overlap and where we differ

Tool type	What it already does	Where we must add value	Position
CRM	Deals, contacts, reminders	Capture from WhatsApp, chase missing info, evidence of completion	Feed or write back to it later; do not replace it
Helpdesk	Tickets, assignment, SLAs	Lightweight setup for firms too small for one	Avoid support-desk positioning
Task tool	Owners, due dates, lists	Auto-create work from inbound messages and follow-ups	Do not become a general task app
Spreadsheet or chat	Free, flexible	Owner, deadline, escalation and report without discipline	Compete on the cost of forgotten requests

Rules to manage the risk

- **Ask in every interview:** what do you use today, and why does it fail on the last ten requests? If the answer is 'nothing fails', disqualify the prospect.
- Our own work-item queue is the system of record only for the covered process. If a customer already runs a CRM or helpdesk, plan a single connector to it, but only after several paying customers request it.
- Never promise a feature a customer's current tool already provides cheaply. Sell the missing step: capture, chase, escalate and prove.
- Add 'we are not a CRM, helpdesk or task app' to the public scope list. Reject feature requests that move us toward being one.

Choose by scored interviews, not assumption

Two candidate segments

- Segment A - mobile and device-repair shops and similar walk-in or drop-off businesses in one Pakistani city. Pain: intake, status questions, pickup follow-up. Reachable through ViroByte's own network. Risk: low price tolerance, much value is status replies that need a reliable status source.
- Segment B - quote-driven service firms: small agencies, maintenance and cleaning firms, trade suppliers. Pain: quote requests and follow-ups that go cold. Higher revenue per job, so a higher fee is more plausible. Risk: harder to reach, and they may already use a CRM.
- Neither is proven. V4 does not assume either.

Scored interview sheet

- Score each segment 1 to 5 on: frequency of the problem, measurable cost of delay, access to buyers, repeatability, supported tools, willingness to pay, low implementation burden. Double-weight willingness to pay and repeatability (maximum 45 points: 5 x 9).
- Record the evidence beside every score. Do not choose from scores alone. A segment must also have a champion, response ownership, account access and permission to measure a baseline.

Interview plan

- Ten interviews in weeks 1 and 2: five per segment. Walk through the last ten requests: where they arrived, which were delayed, who should have acted, how completion was recorded, how often staff chased missing information, which tool already solves part of this, and what would justify a recurring payment.
- Ask what a missed request costs in contribution or staff time. Revenue per job alone is not the value of recovery.

Decision rule

- Pick the segment with the higher weighted score and at least two prospects willing to discuss a paid pilot. If scores are within 4 points, pick the one with the better connector path and buyer access.
- Proceed when three businesses will pay for substantially the same process, one connector path works, and delivery margin is plausible. Otherwise keep interviewing rather than build.
- Defer prospects who want an ERP replacement, bespoke logic per case, unapproved integrations or autonomous high-stakes decisions.

How the same engine serves different businesses

Business types and what changes

Business type	Pack example	What changes
Repair business	Intake and repair-status follow-up	Device, ticket ID, approved status source, collection step
Agency or consultancy	Inquiry and quote follow-through	Project type, budget band, sales owner, quote status
Cleaning or maintenance firm	Service inquiry and scheduling	Service area, site details, duration, resource rules
Wholesaler or distributor	Quotation and order inquiry	SKU references, sales routing, authoritative stock lookup
Professional services firm	Client onboarding checklist	Required documents, reviewer, due dates, confidentiality
Internal operations team	Request and approval tracking	Department, approver, priority, permitted actions

Boundaries by customer size

- **Small businesses:** one workspace and a few operators. Midmarket teams need departments, roles, assignments and reliable integrations. Enterprises may need SSO, provisioning, procurement review and regional controls; these are future funded capabilities, not launch promises.

Compatibility rules

- Adaptability needs accessible source data, a supported process and a tested connector. Physical work still needs a human completion signal; the model never infers that a repair or job is finished.
- Sensitive domains may use administrative intake only after review. Clinical, credit, employment and legal decisions are outside initial scope.

Expansion test

- Before accepting a new industry, show that most steps reuse an existing pack and that no engine fork is needed. Substantial bespoke code is separately priced implementation work, or declined.

Pack catalog and release order

Pack catalog

Pack	Defined result	Release condition
Request Follow-through	Qualified request assigned; next action tracked; closed or escalated with evidence	Only launch pack
Quote Follow-up	Approved quote tracked through reminders to accepted, declined or staff review	Three paying requests; authoritative quote state available
Client Onboarding	Required fields and documents collected and reviewed	Proven file controls and recurring paid need
Service Resolution	Case assigned, status checked, resolution recorded	Reliable ticket or status integration
Booking	Confirmed appointment with conflict handling	Customers require scheduling; resource model tested
Internal Approvals	Request reviewed by the right authorized person	Permission model and approval expiry validated

A pack is a product unit

- Each pack includes a versioned field schema, completion contract, fixed workflow, allowed actions, knowledge starter, policy settings, sample data, failure tests, report definitions and a support playbook.
- Pricing covers an active pack in a workspace, not an undefined promise of automation.

Start thin

- Implement one concrete pack and only the abstractions it needs. Add a reusable interface when a second paid use case shows the commonality. Do not build a universal workflow language before serving customers.

Later capabilities

- A bounded step editor and AI-assisted configuration only after repeated configuration requests. AI may draft supported mappings and rules; owners review, simulate and publish. No arbitrary generated production code and no unrestricted computer-use agent in the core product.

Scope of the first release

Included

- One organization, one inbound channel, one language, the Request Follow-through pack. Default to WhatsApp text if pilot customers use it and onboarding is feasible; otherwise a hosted intake form. Build one live channel behind a common event interface.
- **Intake:** capture contact details and business-defined fields, ask a limited number of clarification questions, create one work item.
- **Follow-through:** assign an owner, set a due date, detect missing information, schedule approved reminders or internal escalation.
- **Human control:** take over, approve an external message, change ownership, resolve an exception, or close with a reason and evidence.
- **Visibility:** work queue, item timeline, connection status, usage statement and weekly outcome report. A simple configuration form, not a visual editor.
- **Operations:** audit trail, tenant isolation, action receipts, retries, backup restore, export, deletion and manual billing records.

Explicitly deferred

- Calendar booking, Google Sheets mirroring, inbound mailbox access, CRM write-back, document extraction, voice, marketplace, resellers and self-service billing. Add one only when the paid pilot's agreed result requires it. The internal database is enough for tracking.

Completion rule

- A request is complete only when its configured evidence exists: operator-confirmed qualification and assignment, a recorded customer decision, or a provider-confirmed ticket result. A sent reminder is an activity, not a completed sale or solved case.

Paid readiness

- A business can activate a tested configuration, handle real requests, pause automation, recover uncertain actions, inspect results and cancel. Provider access must work before charging for a live capability.

Three MVP workflows with examples

W1: intake and assignment

- **Trigger:** a verified inbound event. Resolve tenant and contact; check for an open item with configured matching rules. Extract only supported fields, validate them, ask at most two clarification rounds. Save the request and assign an available owner or a visible unassigned queue. Done when a saved record and a traceable assignment event exist.

W2: due-date follow-through

- **Trigger:** a persisted timer carrying the expected item version. Reload the latest state, recipient, consent, message-window rule, approval and budget. If information is still missing, send the approved request or alert staff.
- **Example pilot policy:** up to two reminders over seven days, then human review. Timing, quiet hours and caps are business-approved settings, not universal defaults.

W3: exception and closure

- **Trigger:** ambiguity, complaint, unsupported action, expired deadline or uncertain provider result. Pause the affected automatic step, assign an operator, show source evidence and a proposed next action. Close only after an authorized operator or trusted system supplies the evidence. No response becomes dormant or escalated, never assumed successful.

Cross-industry example

- An agency receives a project inquiry: the pack collects service type and deadline, assigns a salesperson and tracks whether the next action happens. A repair shop uses the same intake and timer logic with a device type and ticket owner. A 'repair completed' message additionally needs a reliable status source.
- **Roman Urdu sample:** a customer writes 'bhai kal tak phone ready ho jaye ga?'. The system extracts the ticket reference if present, checks the approved status source, and replies only with a confirmed status. If none exists, it hands off to a person.

Acceptance examples

- Replaying an inbound event creates no second request. A customer reply cancels stale reminders. A staff takeover blocks queued AI sends. An unknown write result reaches review before a retry. Reopening a request keeps its history and does not inflate completion counts.

Guided setup, screens and roles

Six setup steps

- 1. Choose a supported process and define the result. 2. Enter business hours, timezone, language and required fields. 3. Connect the channel and verify destinations. 4. Assign owners and define deadlines, approval rules and reminder limits. 5. Run sample normal and failure cases. 6. Review permissions, usage caps and the completion contract, then activate.
- Save after every step. Show missing prerequisites, test versus live mode, destination previews and clear rollback.

Five primary views

- Work queue, item timeline, approvals and exceptions, connections, results and usage. The queue puts overdue, unassigned and blocked work first. Each item shows owner, deadline, latest evidence and next action. Raw workflow IDs and provider errors stay in a troubleshooting panel.

Roles

- **Owner:** billing, access, policies, publishing. Manager: supervises work and approved configuration. Operator: handles assigned requests and exceptions. Support: technical metadata by default; temporary customer-content access is explicitly approved and audited.

Design rules

- Make manual and automated actions visually distinct, and show human confirmation separately from provider confirmation.
- Mobile-friendly layouts, accessible labels and keyboard navigation. Add right-to-left presentation for Urdu only after language-specific QA.

Configuration contract

- Store workspace identity, locale, process version, required fields, allowed destinations, escalation roster, deadline rules, quiet hours, knowledge version, consent rules, approval policy, budget and retention settings. Validate references and permissions before publishing an immutable version.

Onboarding target

- No more than two founder hours for a standard setup after ten similar customers. Track provider-approval wait separately from hands-on effort. Do not advertise the target until observed.

Proving value with the ledger

Evidence ledger

- Record work-item ID, event type, source message or provider ID, actor, timestamp, workflow version and verification level: system-observed, provider-confirmed, customer-reported or operator-confirmed. Keep only what the tenant's retention policy allows.

Weekly report

- Eligible requests, completed items, overdue items, escalation rate, median handling time, manual intervention minutes and delivery cost. Each number links to underlying items. Excluded and unsupported requests are shown separately so the denominator cannot hide poor fit.

Net time saved

- Net time saved = comparable baseline handling time minus current human handling, review and correction time. Include all retained manual steps and allow a negative result.
- **Example:** 100 comparable items at a 6-minute baseline versus 2 minutes of current human work save 400 minutes, or 6.67 hours.

Outcomes are associations

- Accepted quotes and completed jobs count only from an authoritative record or explicit customer confirmation. They are associated outcomes, not automatically revenue caused by the product.
- Compare matched periods and disclose changes in volume, seasonality and staffing. Use a holdout or phased rollout when feasible.

A useful report drives action

- List the three most common reasons items stalled and a proposed configuration improvement. Changes stay drafts until reviewed, tested and published.
- **Value proposition test:** seek a customer-accepted, conservative benefit of about three times the fee as a discovery target, not a guarantee. Section 14 applies it with local labor rates.

Global tier and local tier

Commercial model

- One-time setup fee plus a recurring workspace-and-pack subscription, with a stated item allowance, AI budget and support allowance. Messaging charges are itemized at agreed pass-through cost.
- All prices are planning hypotheses for interviews, not validated market prices. PKR figures assume prices a small Pakistani business could plausibly pay and must be replaced by interview findings.

Global tier (USD)

Offer	Price	Scope and caps
Four-week assisted pilot	$300 setup + $199 pilot fee	One pack; 150 work items; one channel; two hours support
Core	$299 per month + $500 setup	One pack; 300 items; one channel; five users; 30 minutes support
Team - later	$699 per month + scoped setup	Up to three proven packs; 1,000 items; ten users; one hour support

Local tier (PKR)

Offer	Price	Scope and caps
Four-week assisted pilot	PKR 10,000 setup + PKR 8,000 pilot fee	One pack; 150 work items; one channel; two hours support
Local Core	PKR 12,000 per month + PKR 20,000 setup	One pack; 300 items; one channel; three users; 30 minutes support
Local Team - later	PKR 30,000 per month + scoped setup	Up to three proven packs; 1,000 items; five users; one hour support

Billing unit and margin protection

- A work item is one deduplicated request admitted to an enabled pack. Retries, internal steps and reopening the same request are not billable. Define genuine new requests and merges in the contract.
- Internally cap tokens, actions, attachments, lifetime and reminders per item. Alert at 80% of usage or spend; at the cap, pause new automatic work visibly and route it to staff. No unannounced overages, unlimited plans or success fees on unverifiable revenue.
- Pilots are invoiced manually. Choose the tier by the buyer's market, never by assuming location.

Contribution and break-even for both tiers

Assumption notice

- All figures are assumptions for sensitivity analysis, not vendor quotes. Messaging pass-through is excluded from revenue and from the costs shown; count any unrecovered messaging cost in real account costs.

Monthly Core account, global tier (USD)

Item	Base case	High-support case
Subscription revenue	$299	$299
AI and execution	$20	$35
Hosting and email	$12	$15
Payments and refund allowance	$12	$12
Direct support at $30/hour	$15 (0.5 h)	$120 (4 h)
Total variable cost	$59	$182
Contribution	$240 / 80.3%	$117 / 39.1%

Monthly Local Core account (PKR)

Item	Base case	High-support case
Subscription revenue	PKR 12,000	PKR 12,000
AI and execution	1,800	3,000
Hosting	1,000	1,200
Payments and bank charges	400	400
Direct support at PKR 600/hour	300 (0.5 h)	2,400 (4 h)
Total variable cost	3,500	7,000
Contribution	PKR 8,500 / 70.8%	PKR 5,000 / 41.7%

Break-even for a three-founder team

- **Fixed cost assumption:** $600 overhead plus a $3,000 team stipend target, or $3,600 per month. USD tier: 15 accounts at $240 cover it; 20 accounts leave $1,200; at $117 contribution, 31 accounts are needed.
- **Local fixed cost assumption:** PKR 60,000 overhead plus three stipends of PKR 50,000, or PKR 210,000. At PKR 8,500, that is 25 accounts; at PKR 5,000, 42 accounts. The local tier alone reaches founder pay slowly. Treat it as a way to learn and win early customers, and use the USD tier or higher local prices to fund the team.

Formulas and gates

- Contribution = recurring revenue minus variable delivery cost including direct support. Break-even accounts = round up fixed monthly cost divided by contribution. Payback = acquisition cost divided by monthly contribution; $480 over $240 is two months before churn.
- Require positive contribution in pilots and target 70% or more for standardized accounts before paid acquisition. A $500 setup fee yields $200 after ten hours at $30; a PKR 20,000 setup yields PKR 14,000 after ten hours at PKR 600. Track real hours and reprice work above allowance. Contribution is not net profit; model churn and cash separately.

Does the benefit beat the fee at local labor rates

The test

- A buyer should accept a conservative benefit of about three times the monthly fee. Test it twice: with time savings alone, and with recovered requests that would otherwise have been lost. Use the customer's own staff wage, not a foreign one.

Local Core, PKR 12,000 per month (assumed staff cost PKR 400/hour)

Step	Calculation	Result
Target benefit	3 x PKR 12,000	PKR 36,000
Time saved only	100 items x 4 min = 400 min = 6.67 h x PKR 400	PKR 2,667 (0.22x fee)
Time saved at 300 items	3 x 2,667	PKR 8,000 (0.67x fee)
Recovered jobs needed	36,000 / PKR 2,500 contribution per job	14.4, so 15 jobs
As share of allowance	15 of 300 items	5%

Core, $299 per month (assumed staff cost $15/hour)

Step	Calculation	Result
Target benefit	3 x $299	$897
Time saved only	6.67 h x $15 (100 items)	$100 (0.33x fee)
Recovered jobs needed	$897 / $60 contribution per job	14.95, so 15 jobs

What the result means

- At local labor rates, time saved never justifies the fee. The product sells only if it recovers requests that would have been lost, about 15 per month at either tier under these assumptions, or if it prevents costly delays. Confirm the contribution per job and the baseline loss rate in interviews; a customer who loses fewer than about 5% of requests is a poor fit.
- Count a recovered job only when the baseline period shows comparable requests being lost and the ledger shows the item completed. Do not claim recovery from reminders sent.
- If the buyer cannot show enough loss, change the segment, reduce the price or reduce delivery cost. Do not discount first.

Paid pilots and demand validation

Weeks 1 to 4: prove the process

- Interview ten businesses across both segments, choose one, and map a repeated process from authorized examples. Demonstrate intake, follow-up and handoff with labeled sample data. Seek three paid pilots with one agreed result and a working provider path.
- Deliver with existing tools and supervised manual work, and say what is manual. Keep each customer's data separate and check tool licensing before choosing the pilot stack.
- Establish a baseline before activation: volume, handling time, missed or overdue work, ownership and existing costs. Record the sample and gaps.
- Start one customer first. Review failures daily, record manual work and update the shared procedure before the next customer.
- Put support hours, usage, cancellation, refunds for undeliverable setup, customer duties and the ongoing price in writing.

Four-week review gate

- At least three paying customers; at least two willing to renew at the disclosed ongoing price; at least 100 eligible items across the cohort when volume allows; positive contribution per account; a measurable improvement accepted by two customers. If the sample is too small, extend observation instead of declaring success.

Stop or narrow when

- Customers praise demos but will not pay; most requests need bespoke code; no trustworthy completion evidence exists; support work overwhelms the price. Fewer than two renewals from three is an investigation trigger, not a statistical verdict.

Pilot output

- A reusable pack specification, real exception taxonomy, cost record, anonymized test cases and a permission-based case study. Automate the highest-volume repeated manual step first.

Stack and implementation boundaries

Recommended design, subject to checks

- One TypeScript web application for dashboard and API, one long-running worker, Postgres for business state and a Postgres-based job queue (for example pg-boss), and private object storage only when needed. Managed authentication is acceptable. Pin maintained dependencies at build time; this guide is not a version guarantee.

Processing sequence

- Verify the inbound event, resolve tenant, atomically persist event and outbox row, acknowledge receipt, dispatch a durable job, load pinned pack and policy versions, validate extracted data, execute allowed actions, store receipts, then update the work item and report events. Scheduled jobs reload current state before acting.

Principles

- Domain rules are independent of the channel. Adapters normalize inbound messages and expose registered operations with typed inputs and outputs.
- Ordinary code handles deadlines, arithmetic, authorization, consent, state transitions and budgets. AI extracts, classifies and drafts inside those rules.
- Secrets stay server-side. Only the execution service calls external write actions, after policy checks.
- Defer Redis, microservices, a graph editor and a data warehouse until a measured need justifies them.

Repository layout

- apps/web and apps/worker; packages/domain (work-item rules); packages/packs (versioned definitions and acceptance cases); packages/connectors (adapters); packages/ai (prompts and evaluation); db/migrations; tests.

What is original

- Own the completion contracts, work-item lifecycle, exception queue, evidence model and pack tests. Use commodity infrastructure elsewhere. Do not fork or embed a third-party automation platform for customers without checking license and service terms.

Database design and interface contracts

Core tables

- organizations and memberships define tenant access. business_configs and policy_versions hold approved settings. packs and pack_versions are immutable templates. work_items hold org_id, type, status, owner_id, due_at, next_action_at, state_version, policy_version and closure_reason. contacts and channel_identities hold tenant-scoped identities.

Execution and evidence tables

- webhook_events, outbox, workflow_runs, step_runs, scheduled_actions, action_receipts, approvals, evidence_events, outcome_reports, usage_events, subscriptions, audit_events, connections. Every tenant-owned row carries org_id.
- Money in integer minor units with currency; UTC timestamps with an IANA business timezone; schema-validated custom fields.

Keys and approvals

- **Unique inbound key:** provider + account + stable event ID. Unique operation key: tenant + logical action ID. Tenant-safe foreign keys and indexes for open work, due timers and exceptions.
- Approvals store payload hash, recipient, amount if relevant, expiry and policy version. A changed payload invalidates the approval.
- Evidence rows keep source references and verification levels. Deletion follows the retention agreement, and reports must show evidence that has expired.

API surface

- POST /work-items; GET /work-items; POST /work-items/{id}/assign; /close; /reopen; POST /approvals/{id}/decide; POST /packs/{id}/test; /publish; POST /connections/{id}/disconnect; GET /reports/outcomes; GET /usage; POST /exports.

Invariants

- Authorize membership server-side and never trust org_id from a request body alone. Require idempotency keys for visible writes and return a queued operation ID rather than false success. Version event schemas, use stable error codes, publish OpenAPI contracts and test cross-tenant ID substitution.

Runtime, deadlines and failure handling

Separate business and run state

- **Work-item states:** new, awaiting_information, ready, in_progress, awaiting_approval, blocked, completed, cancelled, dormant. Overdue is derived from the deadline, not a status. Run states: queued, running, waiting, succeeded, failed, uncertain, cancelled. A finished technical run is not necessarily a completed business request.

Concurrency and stale actions

- Database locking (advisory locks) plus compare-and-swap on state_version for transitions. Durable action claims and unique operation keys stop two workers running the same action. Never hold a transaction open across a slow provider call. A timer carries its expected item version; a changed item is re-evaluated before sending.

External effects

- Store intended action, request hash and provider response; use provider idempotency where available. After a timeout, reconcile by stable provider identity before retrying. If the provider cannot prove the result, mark it uncertain and assign review. Do not claim universal exactly-once delivery.

Retry and release evidence

- Retry transient failures with bounded backoff and jitter. Permanent auth or destination errors pause the connection and alert staff. Apply per-tenant and per-provider concurrency limits. A dead-letter queue keeps exhausted actions with redacted reasons.
- Reminders recheck completion, customer reply, takeover, opt-out, quiet hours, approval and budget just before dispatch. Once a request is in flight, cancellation may be too late; expose the race and reconcile, never promise takeover reverses a sent message.
- Test worker crash after provider success, duplicate callbacks, out-of-order delivery, two workers claiming one action, late replies and expired approvals. Every unresolved outcome stays visible to an operator.

WhatsApp and optional integrations

One channel first

- Use a provider-supported WhatsApp Business route with tenant-bound sender identities, verified webhooks and delivery callbacks. Compare three routes before choosing: Meta's own Cloud API as a Tech Provider, a global provider such as Twilio, or a regional provider. Check onboarding time, cost, support and eligibility for the actual legal entity.
- Twilio documents a 24-hour customer-service window after the latest inbound customer message; outside it an approved template is required [R4]. Follow consent and opt-out rules. Recheck provider terms before release.

Cost

- Do not model all WhatsApp charges as conversation fees. Twilio describes its own per-message fee ($0.005 per message at the time of checking) plus Meta fees by template or message type [R5]. Actual cost depends on route, market and message category. Quote from the chosen provider and reconcile billed usage.

Connector contract

- Every adapter declares authentication, scopes, operations, field schemas, rate limits, timeouts, idempotency or reconciliation support and a versioned test suite. Missing capabilities show as unsupported. Channel acceptance, delivery and business completion are separate events.

Optional integrations

- Google uses the server-side OAuth flow with scoped permissions, protected tokens and revocation handling [R6]. Sheets is an optional mirror within quota guidance [R8]. Calendar is an optional pack: reserve internally, recheck external availability, create and reconcile the event, then confirm; external edits can still conflict [R7].

Expansion order

- Add the one CRM or ticket system several paying customers request, then a second inbound channel. Email inbox access needs its own scope and review. Cross-channel identity merges need verified linkage or human confirmation, never display-name matching.
- If no supported API exists, offer a labeled manual handoff or decline. Do not offer credential sharing or unattended browser work as a standard connector.

AI behavior and language gates

Bounded responsibilities

- The model classifies intent, extracts required fields, summarizes source messages and drafts replies from approved knowledge. It returns schema-validated fields, evidence references, missing information and a proposed next action. It cannot grant permissions, invent a completion event, set its own budget or change published policy.

Grounding and escalation

- Keep trusted policy, approved business knowledge and untrusted customer content separate. Unknown price, status, availability or policy leads to a question or handoff. Validate extracted IDs, dates and destinations against trusted records. Self-reported model confidence is not authorization.

Evaluation set and gate

- At least 100 representative anonymized or synthetic examples for the launched pack and language: incomplete requests, typos, ambiguous dates, contradictions, complaints, prompt injection, opt-outs and provider failures. Hold out cases and keep human labels.
- **Gate per language:** at least 95% correct extraction of required fields, zero observed unauthorized actions, zero invented material facts. Report numerator, denominator and sample mix.

Language plan

- Evaluate Roman Urdu, Urdu script and English separately, plus mixed-language messages, local dates and numerals. Launch only the languages that pass with competent reviewer coverage. Launch first in the language the paying pilots use. Add Arabic only if a paying customer asks.

Quality beyond accuracy

- Measure unnecessary handoff, missed escalation, latency, cost per eligible item and correction minutes. A high automation rate is worthless if corrections erase the benefit.
- Regression-test prompt, model, knowledge and pack changes and release to one supervised tenant first. Passing tests is not a production accuracy guarantee; keep escalation and sample live runs.

Tenant isolation and enterprise boundaries

Core requirements

- Enforce tenant isolation in APIs, workers, reports, file links and connector bindings. Use row-level security where appropriate and test privileged service paths separately; Supabase documents that service keys can bypass row-level security [R9]. Use least-privilege credentials, encryption, secret rotation and audited support access.
- Verify webhook signatures and reject malformed or oversized payloads. Validate OAuth state and destinations. Recheck authorization before every external write.
- Administrator MFA, role restrictions and approval expiry. Support staff have no standing access to customer conversations.
- Logs hold no credentials and minimal content. Sensitive payloads sit in private storage with retention and deletion schedules.
- Treat retrieved documents and messages as untrusted input. Block arbitrary outbound URLs and private-network access in any future webhook or fetch feature.

Data lifecycle

- Before activation document data categories, purpose, provider processing, retention, export, deletion and backup expiry. Use minimized synthetic data in development. Cross-customer learning uses reviewed abstract rules or explicitly authorized de-identified examples; no pooling of private conversations by default.

Enterprise gate

- Sell SSO, provisioning, granular roles, regional options, audit export, procurement documents, incident process and funded support only after they are built and verified. Do not claim certification or data residency from a cloud region setting alone.

High-stakes boundary

- The product does administrative coordination. It does not autonomously make medical, credit, hiring or legal decisions, move money or issue refunds. Get specialist review before regulated workflows or new jurisdictions. This is a design boundary, not a legal opinion.

Billing, usage, monitoring and support

Manual billing first

- Record plan, period, allowance, entitlement and invoices even when collecting by bank transfer. Confirm payment-provider eligibility for the actual legal entity before building checkout. Later verify billing webhooks and deduplicate events; Stripe documents asynchronous signed delivery [R10].

Usage enforcement

- Keep immutable usage events apart from analytics. Track logical work items, AI spend, provider messages, support minutes and credits. System-caused retries must not inflate billable volume. Show warnings and an explicit policy for pausing or approved overage.

Operational visibility

- Monitor inbound acceptance, queue age, overdue work, uncertain actions, auth failures and cost per tenant. Pilot alerts: oldest ready job above two minutes; sustained action failures above 5% with a meaningful sample; any suspected cross-tenant incident. Tune from real traffic.
- Separate development, staging and production accounts, credentials and data. Release with compatible migrations, feature flags and a tested rollback. Take backups and run restore drills; do not sell uptime or recovery targets before testing.

Support playbooks

- **Revoked access:** pause dependent work and guide reconnection. Uncertain write: inspect receipts before retry. Wrong reply: take over, correct knowledge, add a regression case. Usage spike: inspect loops and pause with notice. Cancellation: stop automation, clarify pending work, export and apply retention.
- Publish support hours and escalation routes. Support on WhatsApp is the default channel for local customers, with response targets stated and kept.

Roles, capacity and timeline

Assumed roles - confirm before starting

- **Founder 1, platform:** architecture, tenant foundation, queue and worker, connectors, reliability. Founder 2, product and AI: dashboard, pack configuration, extraction, evaluation sets, reports. Founder 3, customer: interviews, onboarding, sales, support, billing records and baselines.
- These roles are an assumption based on three co-founders; swap them to match actual strengths. Every phase needs a named owner. Other company work must be ring-fenced, including SchoolDesk time.

Estimated timeline (reduced MVP; provider approvals can extend it)

Phase	Window	Exit evidence
Discovery and concierge pilots	Weeks 1-4	Scored segment decision; three paid commitments; baselines and costs. Founders 1 and 2 build only a throwaway simulator.
Tenant foundation and one adapter	Weeks 5-7	Isolation tests; simulated pack; production access path
End-to-end pack and work queue	Weeks 8-10	Intake, timers, handoff, receipts, evidence working together
Hardening and supervised rollout	Weeks 11-13	Failure tests, restore, usage, export, first live tenant
Renewal and cost validation	Weeks 14-17	Paid renewal, acceptable contribution, support load

Why 17 weeks and not fewer

- Three people shorten the plan by about two weeks against a solo schedule of 19, not by two thirds. Provider approval, pilot observation and the renewal cycle cannot be parallelized away.

Working rules

- Founder 3 reserves 30 to 40% of time for pilots and operations; Founders 1 and 2 reserve about 15% for onboarding and support rotation. If pilots and engineering collide, slow the build or narrow the pilot load. Do not silently create three full-time jobs.
- Hire or contract only when recurring contribution funds the role and the bottleneck is measured. A second pack, second channel or enterprise need gets its own evidence, cost, owner and gate. No fixed date for the broader platform.

Iterations with owners and acceptance

A: tenant foundation (Founder 1)

- Organization and membership models, role checks, audit events, minimal dashboard. Two synthetic tenants attack all initial endpoints. Done: an operator cannot publish a pack or reach another workspace, including through background jobs.

B: configuration and simulation (Founder 2, with 1)

- Fixed field schema, owner assignment, deadline rules, approved knowledge, policy preview, and a fake adapter that records intended actions without sending. Done: a configuration validates, runs normal and exception examples and publishes a version only after review.

C: real intake (Founder 1)

- Signed webhook ingestion, deduplication, outbox, worker, extraction and item creation. Done: repeated delivery makes one work item, source evidence is kept, and invalid fields lead to clarification or handoff.

D: follow-through and operator control (Founders 1 and 2)

- Durable timers, quiet hours, reminder limits, takeover, assignment and closure evidence. Done: replies or closure cancel stale reminders; approval changes block old actions; an operator resolves an uncertain action without replaying the whole workflow.

E: paid operation (Founders 2 and 3)

- Weekly reports, usage, manual entitlements, connection repair, export, cancellation, retention, restore procedure. Done: a pilot runs a full cycle with every reported outcome traceable and support time recorded.

Definition of done for any task

- State user outcome, inputs, permissions, failure behavior, cost impact and acceptance evidence. Review code, run tests, update the support procedure, document rollback. Provider-blocked work is marked blocked, not complete because a simulation passed.

Release gates and acceptance

Required acceptance evidence

Area	Evidence
Tenant security	No cross-tenant read or write through APIs, jobs, reports or files
Event processing	Duplicate and reordered events do not duplicate items or actions
Provider uncertainty	Timeout after possible success triggers reconciliation or visible review
Follow-up correctness	Reply, takeover, closure and opt-out invalidate stale sends
AI behavior	Per-language evaluation passes; unknown facts escalate
Outcome reporting	Counts trace to evidence; reopen and retries do not inflate success
Economics	Cost meter reconciles; caps and customer-approved overage work
Recovery	Worker restart, revoked access and backup restore are rehearsed

Load and usability

- Start with 20 concurrent conversations plus 100 duplicate or reordered events, adjusted to pilot volume. Measure queue delay and provider latency. Watch five representative users run setup and exceptions without coaching. Internal latency goals are not service levels.

Production pilot gate

- No known critical security issue, no unresolved duplicate-action defect, named support ownership, working provider access, validated configuration and a working pause control. Start one tenant in supervised mode and expand after an agreed observation period.

Evidence and stop rules

- Record build ID, schema and pack versions, test dataset version, sample sizes, failures, reviewer and rollout decision. Zero errors in a small set is not proof of zero production risk.
- Freeze expansion on unauthorized actions, invisible lost work, unreliable reporting or support load that breaks contribution targets. Fix the cause before adding connectors.

Founder-led launch and growth playbook

Sell a specific problem

- **Offer:** incoming requests kept assigned and followed up, with a weekly record of completed and overdue work. Tailor examples to the chosen segment. Do not advertise an AI that runs any business or guarantees recovered revenue.

Founder-led acquisition

- Founder 3 contacts 10 to 15 qualified prospects a week through direct outreach, referrals and existing networks, aiming for 3 to 5 conversations. These are activity targets, not conversion forecasts. Show a real process with fictional data, explain setup and total cost, and invite a paid pilot.
- Use ViroByte's Instagram and LinkedIn for proof posts: a before-and-after on a fictional example, then permission-based case studies. Local WhatsApp groups and trade associations are likely more productive than ads.

Discipline

- Track qualified lead, demo, paid pilot, activation, renewal and expansion, with cohort denominators and founder sales time.
- Test two truthful messages (fewer missed follow-ups versus less staff chasing) while keeping service constant.
- Publish a case study only with baseline, dates, sample size, manual effort and limitations. Delay broad paid ads and Product Hunt until onboarding and retention are credible.

Retention and expansion

- Weekly exception reviews make the product part of the team's routine. At renewal show whether outcomes improved and whether the process is still needed. Never hide cancellation, export or history.
- Sell a second pack only for a repeated paid need. Test an adjacent segment after the first pack is stable. Partners come later with explicit onboarding, support ownership, commission and per-client permissions.

Moving across industries and systems

Gate 1: repeatable first pack

- Ten paying customers; two completed billing cycles of retention evidence; contribution after support of 70% or more as a target; falling onboarding effort. Report actual churn and sample size. Revenue growth alone is insufficient if support grows faster.

Gate 2: second pack

- At least three existing or committed customers will pay for the same additional process. Reuse the lifecycle, policy checks and evidence model. Add pack tests, completion definitions and support ownership. Measure connector maintenance before promising bundles.

Gate 3: adjacent industry

- Map the new industry to an existing pack. Aim for 80% or more reusable documented steps and no customer-specific engine fork. This is an internal heuristic, not a measure of market fit.

Gate 4: more channels and systems

- Add one connector only when several customers need it, its API and permissions are feasible, and recurring contribution covers maintenance. Validate identity linking, source-of-truth ownership and receipts across channels.

Gate 5: enterprise and partners

- Require paid discovery or a contracted requirement, funded security and support work and a named owner. Partner access uses explicit per-client membership; template changes never silently reach live customer workflows.

Long-term option

- A curated marketplace and bounded editor become useful only when internal packs and demand justify them. Keep the company's central promise on reliable follow-through.

Experiments and stop rules

Customer value metrics

- Eligible request completion rate, overdue fraction, median time to next action, human handling minutes, correction minutes, recovered requests and customer-confirmed results. Separate unsupported requests, provider failures, customer silence and staff delay.

Company health metrics

- Recurring revenue excluding setup, gross retention, contribution after support, acquisition cost including founder sales time, payback, setup labor, support minutes and cash runway. For tiny cohorts show counts beside percentages.

Experiments

Experiment	Evidence	Decision
Segment fit	Three paid pilots for the same process	Narrow or change if customization dominates
Price acceptance	Renewals at the disclosed price, by tier	Change value or costs before deep discounts
Value test	Recovered requests and net time saved (section 14)	Do not sell savings the process cannot produce
Standardization	Support time and setup effort per cohort	Automate repeated work; charge for exceptions
Acquisition	Fully loaded cost against monthly contribution	Pause channels with implausible payback

Proposed stop rules

- Fewer than two of three pilots renew: investigate before expansion. Support above two hours per standardized account per month for two cycles: narrow scope or move the customer to a priced managed tier. Contribution below 70%: no paid acquisition until corrected. Any unauthorized action: pause the workflow and investigate immediately.
- Record hypothesis, start date, cohort, baseline, outcome and decision. Change one major commercial assumption at a time. Do not declare product-market fit from a handful of pilots.

Risks and mitigations

Risks

Risk	Early signal	Response
Generic positioning	Buyers see a chatbot or CRM replacement	One costly unfinished process; show evidence
Tool overlap	'Our CRM or helpdesk does this'	Sell capture, chase, escalate and prove; connect rather than replace
Service trap	Every account needs custom work	Fixed pack boundary; paid custom scope; no engine forks
Weak economics	Support eats the subscription	Measure time, reprice, narrow or stop
Local price ceiling	Local tier cannot fund the team	Mix tiers; raise value; cut delivery cost
Bad completion evidence	Closed items lack an authoritative source	Separate activity from outcome; require review
Provider dependence	Approval delays, cost changes, revoked access	One adapter, visible degradation, export and fallbacks
Unreliable reminders	Sends after reply or closure	Version checks and dispatch-time revalidation
Data exposure	Tenant mapping or support-access error	Layered controls, review, incident pause
Founder overload	Deadlines slip as support rises	Cap pilots, reserve time, rotate support
Over-expansion	New sectors need many connectors	Expansion gates and per-account profitability

Incident response

- Pause affected actions, retain evidence, assess affected tenants, repair, reconcile uncertain external effects and communicate through the agreed process. Add a regression test and record the release decision. Get professional advice on notification duties for the market.

Commercial honesty

- Do not claim the idea is globally unique or guaranteed profitable. Differentiation may come from execution in a selected process; that needs buyer comparisons, paid retention and delivery economics.

Starting plan and founder decisions

Week 1: choose the problem (Founder 3 leads)

- Interview five owners across the two segments; map ten recent requests per willing business; record tools, delay cost and workarounds. Draft the single-pack completion contract. Founder 1 starts WhatsApp route and provider eligibility checks. Founder 2 prepares the scored interview sheet and the sample data.

Week 2: seek paid commitments

- Finish ten interviews, score both segments and decide. Demonstrate a normal case and a failure case with labeled sample data. Present written pilot scope, both price tiers and the ongoing price. Seek three paid commitments and agree baseline measurement.

Week 3: operate one pilot

- Start the first customer with clearly disclosed assisted delivery. Track each manual step, support minute, provider cost and outcome. Collect permissioned examples for tests. Adjust required fields, escalation and the next-action policy.

Week 4: decide what to automate

- Send an evidence-linked weekly report, discuss the ongoing price and onboard the next customer only after the first is stable. Write the pack acceptance suite and rank the most frequent manual task. Start tenant foundations only if the discovery gate is credible.

Decision log to complete

- D01 segment and country; D02 process and completion evidence; D03 language and timezone; D04 channel and provider; D05 source of truth; D06 price tier, support and caps; D07 legal entity and payment method; D08 data handling and retention; D09 founder roles, staffing and support hours; D10 go or no-go criteria. Record owner, evidence, date and reconsideration trigger for each.

Day-30 deliverable

- One paid operating example, a measured cost model and a testable pack specification. A complete general-purpose SaaS is not the month-one goal.

Glossary, sources and verification notes

Key terms

- **Work item:** a tracked business request. Completion contract: rules and evidence required to finish it. Process pack: tested workflow plus configuration and reporting. Industry preset: vocabulary and default fields for a segment. Evidence ledger: traceable records of actions and results. Contribution: revenue minus variable delivery cost including direct support. Net profit also subtracts fixed costs, compensation and taxes.

Sources, official pages

- **[R1] Respond.io, AI Agents:** https://respond.io/ai-agents (checked 10 Oct 2026). [R2] Zapier Agents: https://zapier.com/agents (checked). [R3] n8n AI documentation: https://docs.n8n.io/advanced-ai/ (checked). [R4] Twilio WhatsApp API: https://www.twilio.com/docs/whatsapp/api (checked). [R5] Twilio WhatsApp pricing: https://www.twilio.com/en-us/whatsapp/pricing (checked).
- [R6] Google OAuth for web server apps: https://developers.google.com/identity/protocols/oauth2/web-server. [R7] Google Calendar events.insert: https://developers.google.com/workspace/calendar/api/v3/reference/events/insert. [R8] Google Sheets usage limits: https://developers.google.com/workspace/sheets/api/limits. [R9] Supabase Row Level Security: https://supabase.com/docs/guides/database/postgres/row-level-security. [R10] Stripe webhooks: https://docs.stripe.com/webhooks. R6 to R10 were carried over from V3 and not re-checked for V4; verify before building.
- **[R11] Grok Bot (xAI):** https://x.ai/bot.

Evidence boundaries

- Official pages show feature overlap, not competitor quality, satisfaction or profitability. The recommended differentiation is an inference to test. No demand, willingness to pay or revenue forecast is asserted. Provider terms, APIs and prices change; pin versions and recheck before launch.

Interpretation

- This is an implementation blueprint, not software, a validated business model or a market study. All prices (USD and PKR), labor rates, schedules, founder roles and thresholds are planning assumptions to be replaced with observed data. V4 merges V2 and V3 and supersedes both.